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MILITARY SPECIFICATION

CRESYLIC ACID, TECHNICAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers technical grade cresylic acid.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

VV–L–791 —Lubricants, Liquid Fuels and Related Products; Methods of Inspection, Sampling and Testing.
LLL–B–631 —Boxes; Fiber Corrugated (For Domestic Shipment).
PPP–D–729 —Drums; Metal, 55-Gallon (For Shipment of Noncorrosive Materials).

MILITARY

JAN–P–106 —Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
MIL–C–124 —Containers, (Cans, Pails and Drums) Metal (For Other Than Subsistence Items).

STANDARDS

MILITARY

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

D850—Methods of Test for Distillation of Industrial Aromatic Hydrocarbons.
D255—Method of Test for Steam Distillation of Bituminous Protective Coating.

(Copies of A.S.T.M. publications may be obtained from American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

CODE OF FEDERAL REGULATIONS
INTERSTATE COMMERCE COMMISSION

49 CFR 71–78—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49" CFR 71–78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Moisture. The moisture content of the material shall not be greater than 1.0 percent when tested as specified in 4.4.1.

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3.2 Distillation range. Cresylic acid which has a distillation range conforming to table I, when tested as specified in 4.4.2, shall be considered acceptable.

TABLE I.—*Distillation range*

Distillation (percent)	At temperature of (°C.)	
	(min.)	(max.)
5.0	180	208
50.0	—	212
95.0	—	225
Dry point	—	230

3.3 Oil and naphthalene. The oil and naphthalene content shall not exceed 0.25 percent when tested as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of the material produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2 Sampling. A representative 1-quart specimen shall be removed from each five representative containers taken from each lot by the Government inspector. Each specimen shall be placed in a separate clean, dry container labeled to identify the specimen with the lot and container represented. Each specimen shall be tested separately as specified in 4.4. If there are fewer than five containers in the lot, a specimen shall be removed from each container.

4.3 Inspection. The inspector shall inspect the packing and marking for compliance with section 5.

4.4 Tests.

4.4.1 *Moisture.* The moisture content shall be determined by method 3001.5 of Specification VV–L–791.

4.4.2 *Distillation range.* The distillation range of the specimen shall be determined by A.S.T.M., method D–850.

4.4.3 *Oil and naphthalene.*

4.4.3.1 *Apparatus.* Assemble a steam distillation apparatus, similar to that used in A.S.T.M. method D255 with the following exceptions:

(1) Insert a steam trap betwen the generator and the distillation flask to insure dry steam.

(2) Use an adapter sufficiently long to reach to the bottom of the separatory funnel.

4.4.3.2 *Procedure.* Measure a 100 milliliter (ml.) sample of the material into a 3-liter short, ring-necked, round-bottom flask. Add 400 ml. of a 1.3 specific gravity solution of sodium hydroxide and agitate thoroughly. Place 50 ml. of a 5-percent sodium hydroxide solution in a 2-liter separatory funnel, and place it under the adapter in such a manner that the orifice of the tube is below the surface of the caustic solution. Connect the steam tube and start the distillation. Continue the distillation until the distillate is colorless, then remove the flask and without disturbing the position of the adapter, rinse the adapter and the condenser several times with ether. Remove the separatory funnel from the adapter and cool to below room temperature, then add 100 ml. of ether and shake thoroughly. Release any pressure caused by the shaking, by inverting the funnel and opening the stop-cock slowly. Place the mixture in a cooling bath until a clear separation of the water and ether is apparent. Draw the water layer off into another 2-liter separatory funnel and again extract with 100 ml. of ether. Combine the ether extracts and wash, first with 20 ml. of 10-percent sodium hydroxide solution and then with 50-ml. portions of water until the wash water is neutral to phenolphthalein. Filter the washed ether and extract into a dry, tared 250-ml. Erlenmeyer flask. Rinse the separator funnel and filter pads several times with fresh ether.

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Place the flask containing the ether, oil, and naphthalene on a steam bath and evaporate the ether (the ether may be recovered by use of a condenser), then place the flask and residue in a 90±2° C. oven for exactly 45 minutes, cool in a desiccator and weigh.

$$\text{Percent oil and naphthalene} = \frac{A}{100B}$$

where:

A=weight of residue, gram (gm.)
B=specific gravity of sample (25°C/15.5°)

4.5 Rejection and resubmission. If any specimen fails to conform to this specification, the lot represented shall be rejected. The contractor, at no expense to the Government, shall have the option of having a partial or complete analysis made on each container in the lot, removing the nonconforming materials, and resubmitting the remaining portion of the lot for acceptance testing. If any specimen taken from the resubmitted portion of the lot fails to conform to this specification, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A, military package.* Not less than eight pounds of cresylic acid shall be packaged in a one-gallon type I metal container provided with a screw-cap, inner seal and a formed bridgetype handle, conforming to Federal Specification PPP–C–96.

5.1.2 *Level B, limited military package.* Same as 5.1.1.

5.1.3 *Level C, minimum military package.* Same as 5.1.1.

5.2 Packing.

5.2.1 *Level A, military pack.*

5.2.1.1 *Drums.* Not less than four hundred and forty-pounds of material shall be packed in a type II metal drum conforming to Specification PPP–D–729.

5.2.1.2 *One-gallon containers.* A uniform number of containers of material packaged as specified in 5.1.1 shall be packed in a snug-fitting box, style RSC, compliance symbol V3s, conforming to Specification JAN–P–108, and shall be closed and sealed in accordance with the appendix thereto. Each pack shall be overpacked in a snug-fitting nailed wood box, style 2, 2½, or 4, conforming to Specification JAN–B–106. The box shall be bound with flat metal strapping in conformance with the appendix to Specification JAN–P–106.

5.2.2 *Level B, limited military pack.*

5.2.2.1 *Drums.* Same as 5.2.1.1.

5.2.2.2 *One-gallon containers.* A uniform number of containers of material packaged as specified in 5.1.1 shall be packed in a snug-fitting box, style RSC, having lapped joints and conforming to Specification LLL–B–631. The box shall be closed in conformance with Specification LLL–B–631.

5.2.3 *Level C, minimum military pack.*

5.2.3.1 *Tank cars and tank trucks.* When specified, the material shall be shipped in tank cars or tank trucks meeting the requirements of I.C.C. regulations.

5.2.3.2 *Drums.* Same as specified in 5.2.1.1.

5.2.3.3. *One-gallon containers.* Same as specified in 5.2.2.2.

5.3 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The cresylic acid specified herein is intended for use as a peptizing agent in incendiary oils and flame thrower fuels.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Type of packing required (see 5.1 and 5.2).

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6.3 **Batch.** A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 **Sampling and testing.** When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government at its discretion, may modify in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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